Quantification: 876_z82c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	7
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 7)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1894.71	1908	652.965	px^2
major_axis_length_px	lacuna shape	78.133	83.3807	19.1736	px
minor_axis_length_px	lacuna shape	31.3292	32.7068	5.961	px
aspect_ratio	lacuna shape	2.5011	2.5035	0.4601	unitless
eccentricity	lacuna shape	0.9069	0.9168	0.0417	unitless
solidity	lacuna shape	0.8501	0.8645	0.0414	unitless
roots_count	headline	10.4286	11	4.2762	count
ring_length_r30_px	headline	622.857	637	120.475	px
ring_length_r60_px	reported	1701.57	1820	373.067	px
owned_length_px	ownership-dependent	7582.61	8241.08	3487.96	px
edge_count	ownership-dependent	557.429	618	256.793	count
mean_edge_length_px	ownership-dependent	13.5568	13.3351	0.586	px
perimeter_px	normalised	244.779	259.262	63.1398	px
ring_area_r30_px2	normalised	8695.43	9001	1449.19	px^2
ring_area_r60_px2	normalised	22533.3	23515	3595.4	px^2
in_frame_fraction_r30	normalised	0.9985	1	0.0039	unitless
in_frame_fraction_r60	normalised	0.9791	1	0.0552	unitless
ring_density_r30	normalised	0.0718519	0.0711589	0.009634	px^-1
ring_density_r60	normalised	0.0754602	0.0780355	0.011188	px^-1
roots_per_100px_perimeter	normalised	4.1742	4.6285	0.9979	per 100 px
ring_attached_length_r30_px	network v2	566.429	588	107.14	px
ring_attached_length_r60_px	network v2	1490.86	1663	385.653	px
ring_length_w_r30_px	network v2	700.45	723.973	131.679	px
ring_length_w_r60_px	network v2	1914.01	2069.25	418.36	px
sholl_crossings_r10	network v2	14.8571	16	3.2367	count
sholl_crossings_r20	network v2	18.8571	20	4.4508	count
sholl_crossings_r30	network v2	21.1429	21	3.132	count
loop_count	loops, report only	92.5714	101	54.8114	count
loop_edge_share	loops, report only	0.5297	0.6221	0.1422	share
ring_width_mean_r30_px	width	6.5345	6.7465	0.7076	px

Per field
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measure	value	units
analysed_area_px2	1.0313e+06	px^2
total_skeleton_length_px	70909	px
canalicular_length_density_per_px	0.0687569	px^-1
junction_count	2549	count
junction_density_per_px2	0.00247164	px^-2
lacuna_count	8	count

lacunae_per_px2	7.76e-06	px^-2
skeleton_component_count	628	count
mean_component_length_px	112.912	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0773151	px^-1
field_density_without_flagged_per_px	0.0684623	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	147	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.0687569	px^-1
field_length_density_with_unattached_per_px	0.0688995	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	794	count
loop_edge_share	0.4932	share
edge_count	5507	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.228	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	57437	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4013	213.469	32.7614	6.5158	0.9882	0.5369	1.527	16.3571	878.108	True
2	2473	101.199	32.7068	3.0941	0.9463	0.7952	1.5403	149.487	378.777	False
3	2634	93.9252	37.4915	2.5052	0.9169	0.8299	-1.4941	300.788	911.93	False
4	2169	83.3807	34.4381	2.4212	0.9107	0.8757	-1.5585	303.736	321.658	False
5	1583	72.1147	28.8056	2.5035	0.9168	0.876	-1.5035	531.782	196.779	False
6	1826	64.8209	36.821	1.7604	0.823	0.9062	-1.5669	541.512	442.987	False
7	1908	86.6787	28.6695	3.0234	0.9437	0.8645	-1.4722	691.032	607.199	False
8	670	44.8117	20.3719	2.1997	0.8907	0.8034	-1.4095	908.587	49.1806	False

Network per lacuna

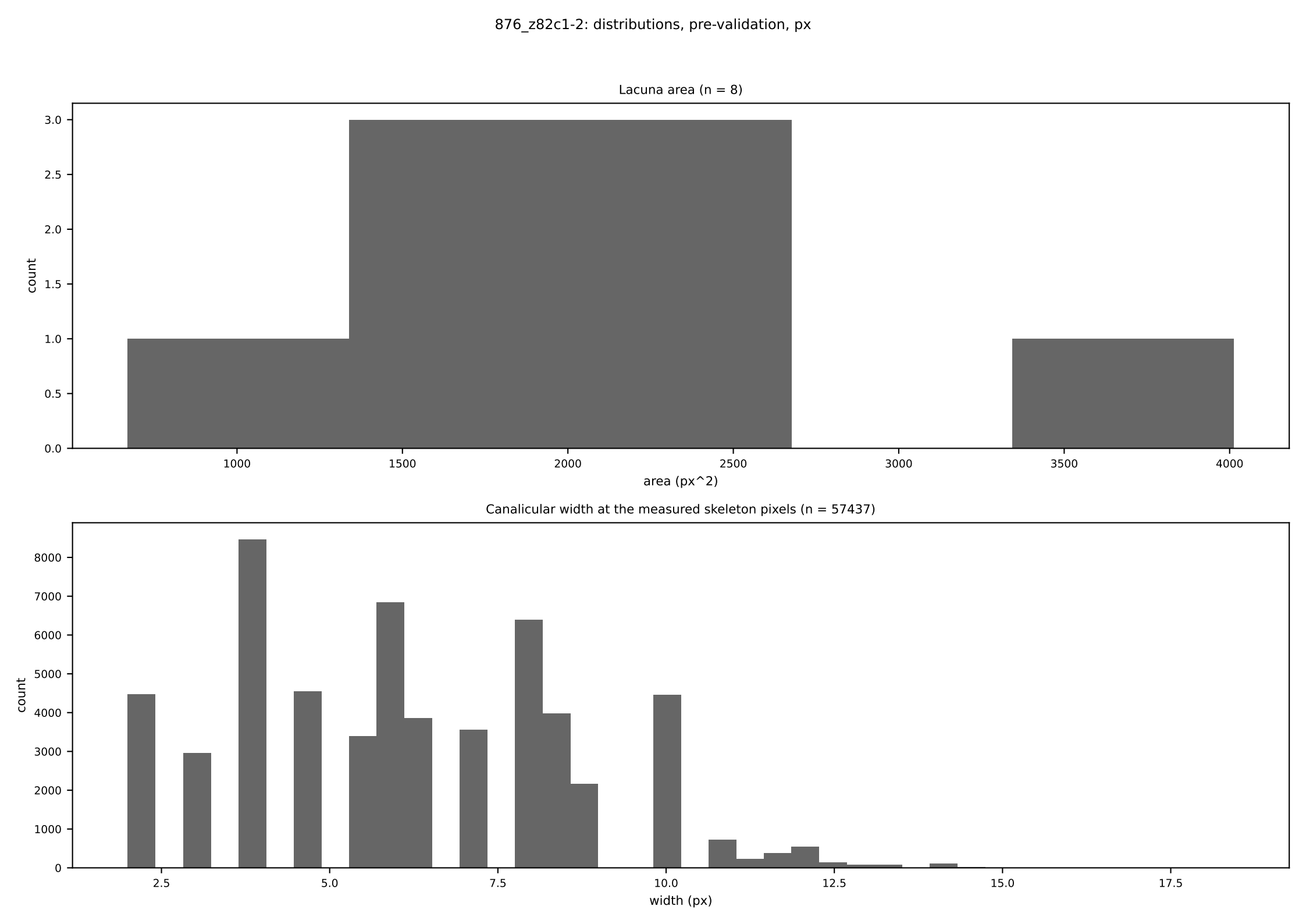
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	26	1091	2076	5257.3	377	13.9451	3
2	15	735	2029	10197.9	696	14.6521	7.6575
3	16	672	1942	11732.3	886	13.2419	6.1937
4	12	757	1961	8241.08	618	13.3351	6.7465
5	11	637	1818	10037.6	756	13.2772	6.8482
6	5	631	1820	6573.25	479	13.7229	6.8296
7	8	505	1245	4179.11	302	13.8381	5.4964
8	6	423	1096	2117.05	165	12.8306	5.9698

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	782.839	12986	24751	0.6542	0.5574	0.0840136	0.0838754	3.3212
2	318.718	10329	26001	1	1	0.0711589	0.0780355	4.7064
3	314.274	10290	25964	1	1	0.0653061	0.0747959	5.0911
4	259.262	9043	23619	1	1	0.0837112	0.0830264	4.6285
5	218.836	8099	21764	1	1	0.0786517	0.0835324	5.0266
6	196.936	7897	21388	1	1	0.0799038	0.0850945	2.5389
7	261.505	9001	23515	1	1	0.0561049	0.0529449	3.0592
8	143.924	6209	15482	0.9898	0.854	0.0681269	0.0707919	4.1689

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	1054	1998	1221.34	2317.26	24	26	17
2	698	1770	820.056	2282.39	19	22	23
3	514	1663	738.073	2154.01	17	21	23
4	681	1837	850.843	2202.63	13	24	26
5	605	1681	723.973	2059.42	12	19	21
6	588	1610	719.6	2069.25	17	20	18
7	467	971	566.688	1389.92	16	15	17
8	412	904	483.919	1240.49	10	11	20


876_z82c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 57437)
count
width (px)